AURAMCP: Enhancing nanoHUB's FAIR Workflows and Data through MCP Servers

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Introduction

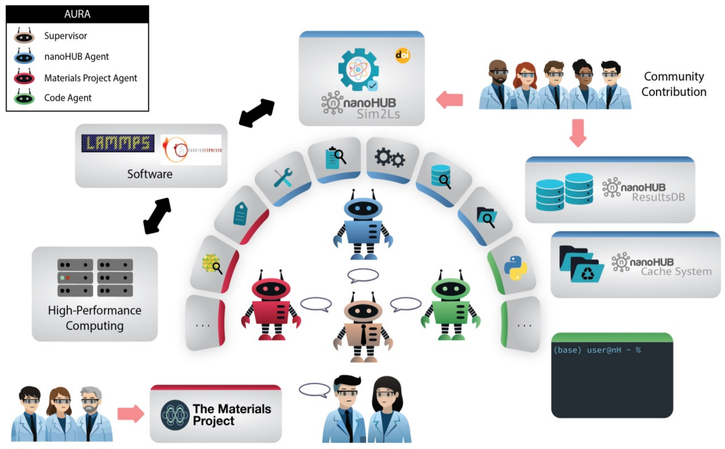
AI agents have become powerful drivers of autonomous scientific workflows[1], capable of adaptive reasoning, tool use, and multi-step planning that goes far beyond single-prompt queries.

AURA[2] is a multi-agent system capable of finding and performing simulations based on user instructions on a task; it also serves as an orchestrator to compose scientific workflows.

But AURA's capability is locked in. It was model-specific and closed to outside systems, meaning that speed and orchestration didn't translate into portability.

AURA: Autonomous Universal Research Assistant

Multi-agent structure: Supervisor + three specialized agents



Key innovation: tools are discovered via metadata at runtime, not hardcoded toolsets → agents figure out *what's available* rather than being told in advance.

Built on a custom LangGraph implementation, with prompts tuned specifically for GPT5.2

References & Acknowledgements

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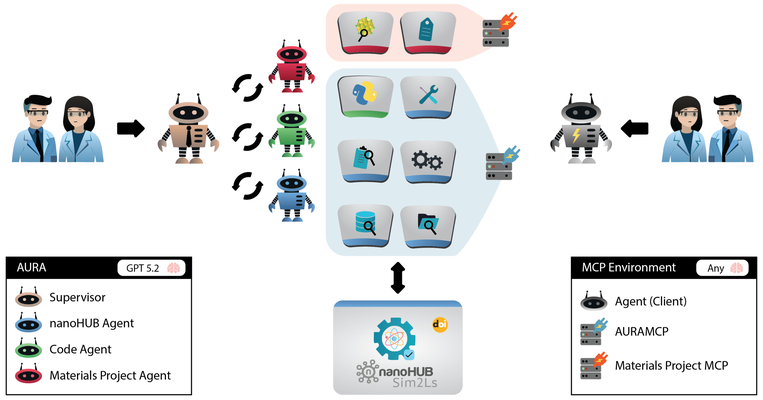
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2. Verduzco, J. C., et al. (2026). Autonomous Universal Research Assistant (AURA): Agentic AI meets nanoHUB's FAIR Workflows and Data.
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AURAMCP: Making AURA's Tools Interoperable

Model-Context-Protocol[3] (MCP) is an open standard that lets AI agents discover and call external tools through a shared, predictable interface. It serves like a USB port for AI, so any compatible agent can plug into any compatible tool without custom integration work.

BEFORE
X Not reusable by other agents
X Every new model requires re-prompting → cumbersome

AFTER
✓ Callable by any MCP client
✓ Discoverable, versioned, reusable → user freedom



Sim2L[4] packages a nanoHUB simulation into a callable, versioned unit with defined inputs and outputs. AURAMCP exposes internal tools to query and execute Sim2Ls, ultimately producing the results the user requested.

	Input	Output
Materials Melting Point	**Alloy Composition**	**Melting Temperature (e.g. 1891.7K)**
Protein Shape	**Amino Acid Sequence**	**3D Structure Visual (pLDDT 90-98)**
Elastic constants	**Structure + DFT Parameters**	**Elastic Tensor (e.g. C_{11} = 820.7 GPa)**

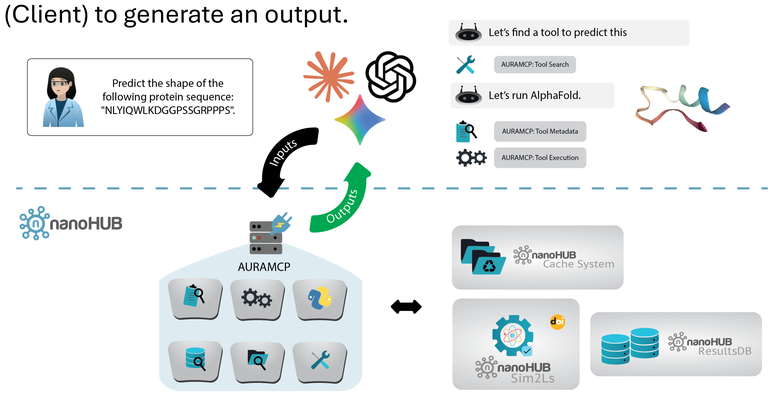
Quality of Life Improvements & Tracing

- **Cleaner outputs.** Takes advantage of visualization tools
- **Fewer roadblocks.** Workflows hit noticeably fewer errors.
- **Real-time visibility.** Able to redirect a task mid-run.
- **Graceful pausing.** Claude's pause behavior kept workflows alive rather than killing them outright, avoiding full restarts.
- **Direct Tracing.** Claude's responses made it clear *which* MCP or tool was failing (server-side vs. tool-specific vs. connection)

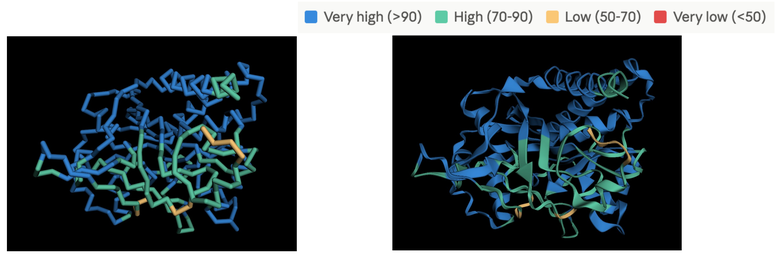
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Capabilities / Example Workflows

AURAMCP enables users without any domain knowledge to run a research-grade workflow simply by collaborating with the Agent (Client) to generate an output.

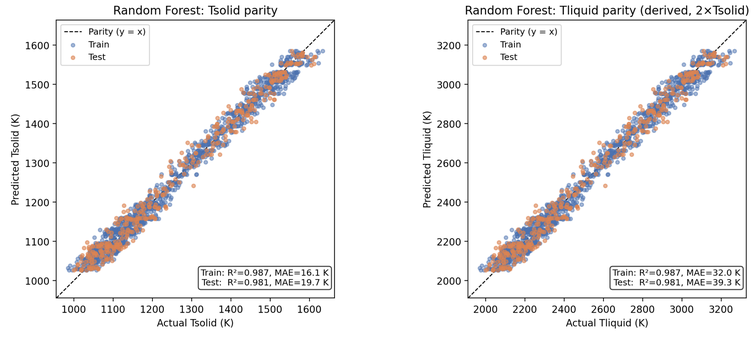


"Predict the shape of the following protein sequence: (Sequence for Asparagine Synthetase)." (ESMFold)



Backbone format | Ribbons and Segments

"Predict the melting temperature of a high-entropy alloy." (MeltHEAS)



Conclusions

✓ AURAMCP brings interoperability and accessibility to clients (Agents) beyond AURA's original implementation.
✓ Validated across three simulation families: Melting point predictions, protein structure prediction, and elastic calculations.
✓ The pattern generalizes: any nanoHUB Sim2L can be discovered and executed as a tool, callable by any MCP-compatible agent (e.g. Opus 4.8, GPT 5.5) .